

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

NEWTON'S LAWS OF MOTION

LECTURE-2

PHYSICS

WAJAHID SIR (Gabbar Sir)



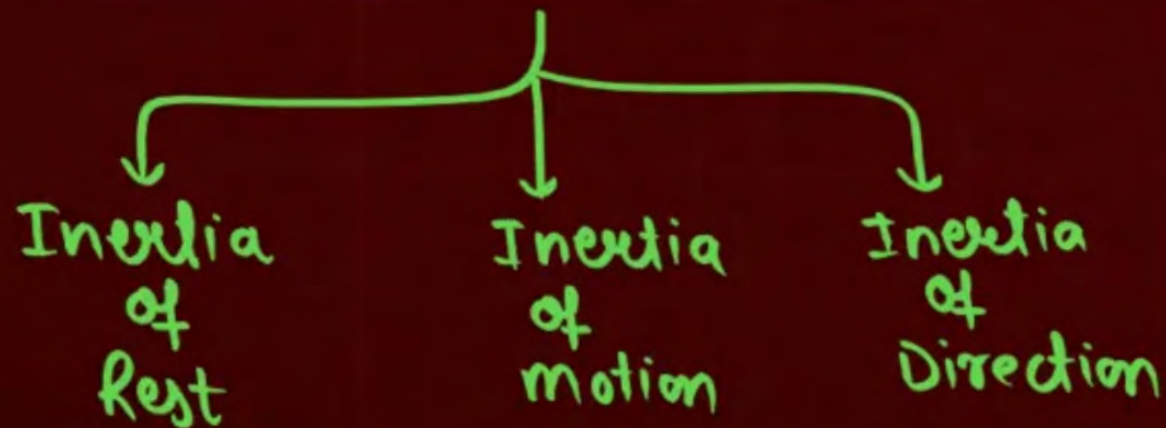
Topics To Be Covered

- 1 Revision ✓
- 2 Newton 3rd Law of Motion ✓
- 3 Law of Conservation of Momentum ✓
- 4 Application of Law of Conservation of Linear Momentum

Recap From 1st Lecture

Inertia (1st law of motion)

Property of Body by virtue of which Body cannot change its state of Rest, motion and Direction



Momentum (p)

$$\vec{p} = m\vec{v} \quad \text{kg m/s vector}$$

$$\bullet p_f = mv \quad \bullet p_i = mv$$

② # Newton 2nd laws of Motion

$$F = \frac{\Delta p}{\Delta t} = ma \rightarrow a^{\text{th}}$$

$$F = \frac{\Delta p}{\Delta t} = \frac{dp}{dt} = m \frac{dv}{dt} + v \frac{dm}{dt}$$

③ Impulse

Force acting for the short period of time.

$$\uparrow F = \frac{\Delta p}{\Delta t} \downarrow$$

$$F = \frac{dp}{dt}$$

$$\underbrace{F dt}_{\text{Impulse}} = dp$$

Impulse momentum theorem

Newton 3rd Law of Motion

lecture.no.②



Every action has an equal and opposite reaction.

Action = - Reaction
↓ ↓
Force Force

Forces always exist in pairs.

Both the force are of same nature

Both the force act on different Body



Application of 3rd law of motion

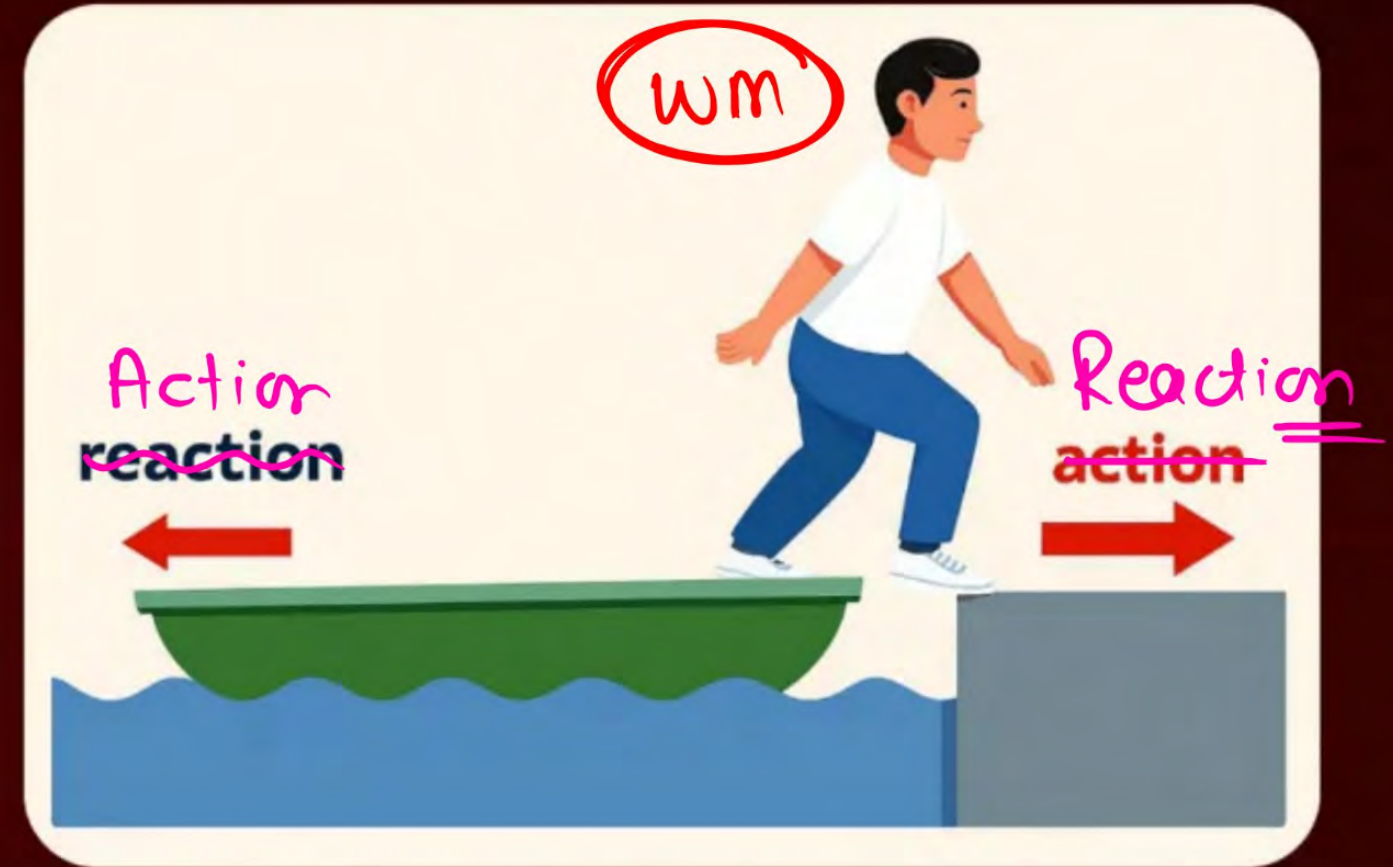
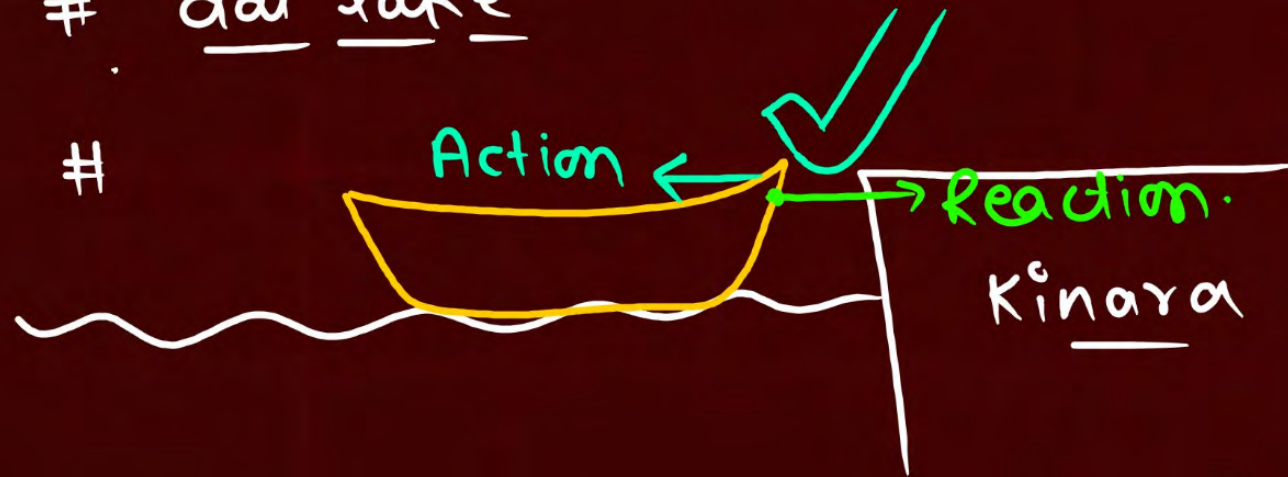
1

Daka from Boat

Kashmir

dal lake

#



2 Rocket Propulsion

My Best friend (Bittu)

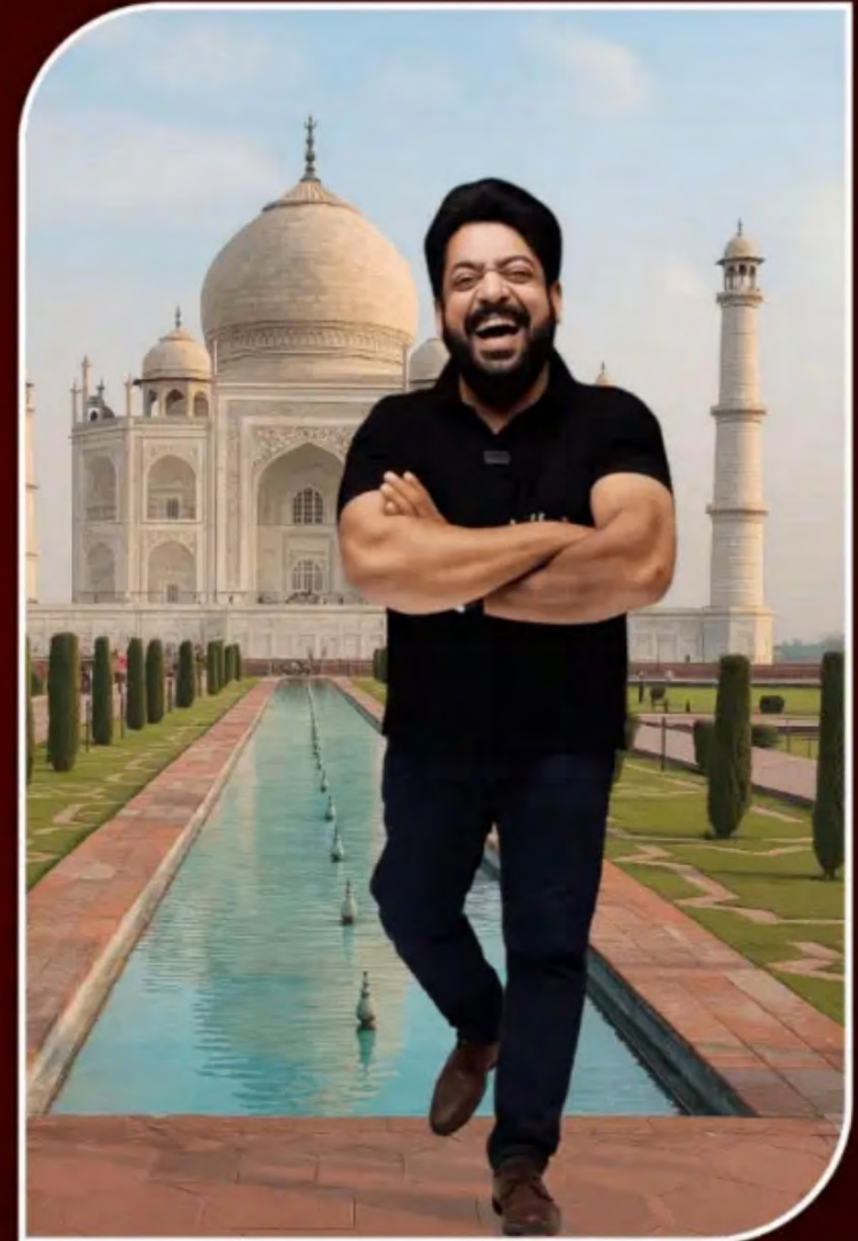
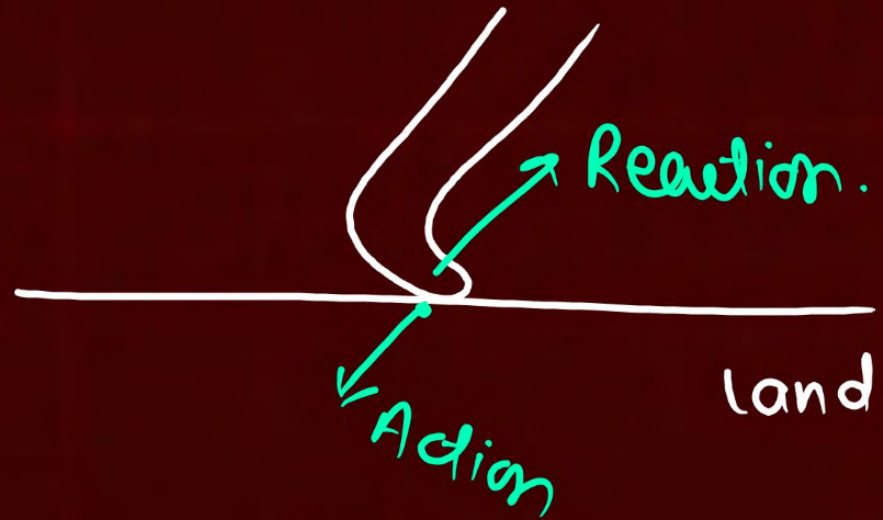
ISRO (Kerala)



3 Walking

EK din

Master visit the Taj-Mahal.



4

Boat



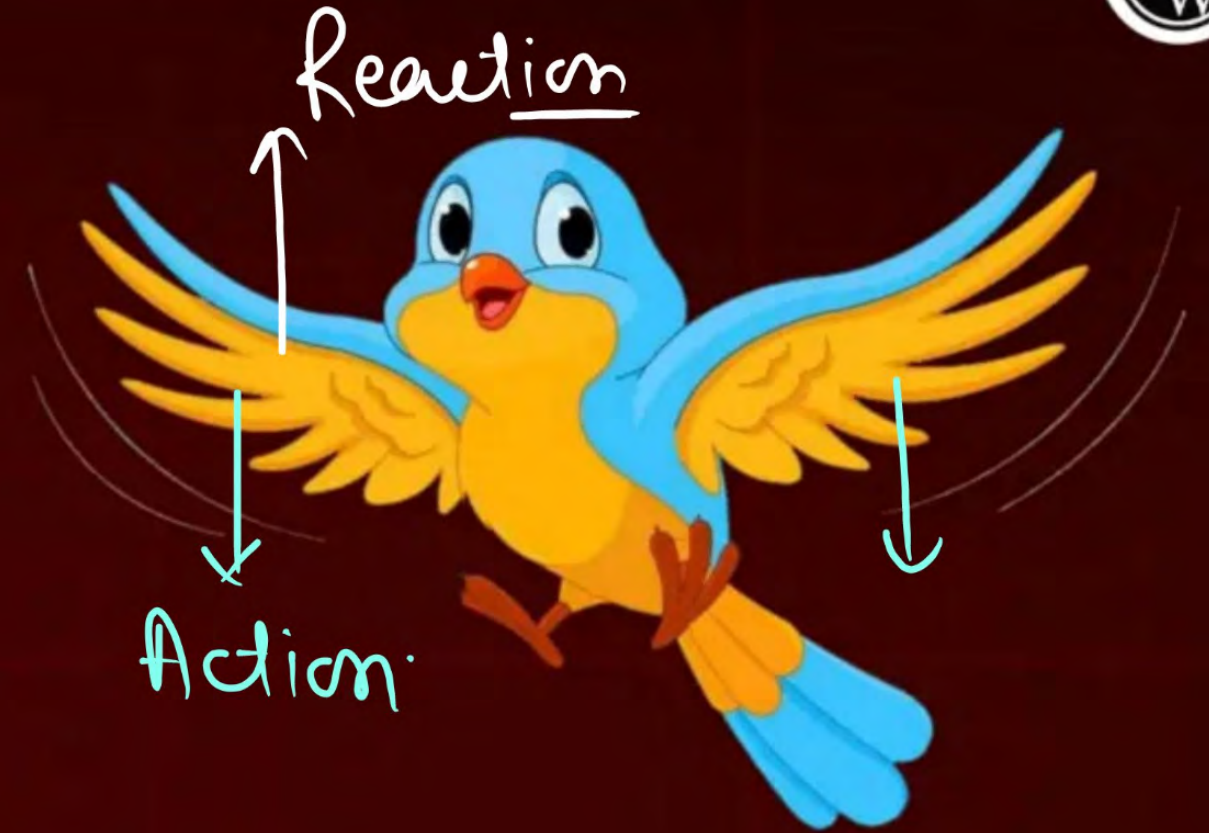
Sailing of Boat.

mca - 2024



5 Flying of Birds

Parrot (saritaj)



6

Imp.

Sun Shot

Mamu (Ghani Uncle)



←
Action



→ Reaction

Law of Conservation of Momentum

③ mark

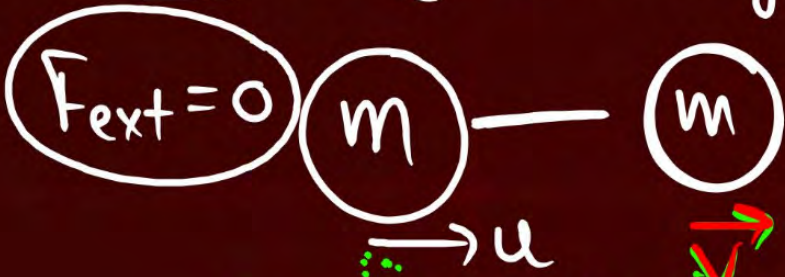
2024
2023

2022



It states that "If the net external force on a system is zero, then the total linear momentum of the isolated system will always remain conserved". ($P_i = P_f$)

Case ① (one Body system)



$$F_{ext} = \frac{\Delta p}{\Delta t}$$

$$0 = \frac{\Delta p}{\Delta t}$$

$$\Delta t \times 0 = \Delta p$$

$$0 = \Delta p$$

$$\Delta p = 0$$

$$p_f - p_i = 0$$

$$p_f = p_i$$

$$p_f = p_i$$

$$\cancel{m}v = \cancel{m}u$$

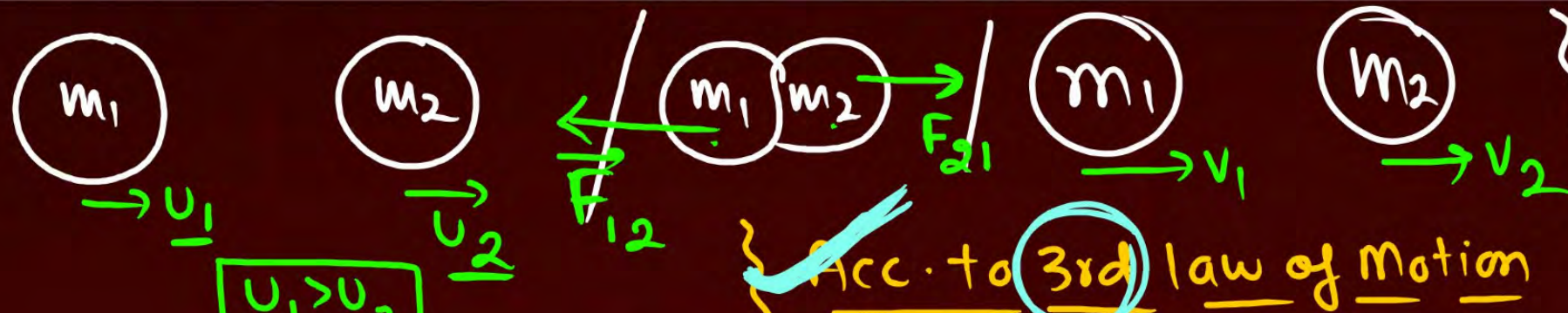
$$v = u$$

$$\underline{2\text{ m/s}} \quad \underline{2\text{ m/s}}$$



Case 2

Momentum Conservation for two bodies



Phella hi dayaan Rakhna hai

$$P_{i1} = m_1 u_1$$

$$P_{i2} = m_2 u_2$$

$$P_{f1} = m_1 v_1 \checkmark$$

$$P_{f2} = m_2 v_2$$

$$F = \frac{\Delta P}{\Delta t}$$

Acc. to 3rd law of motion

Action = - Reaction

$$\vec{F}_{21} = -\vec{F}_{12}$$

$$\vec{f}_2 = -\vec{f}_1$$

$$\frac{\Delta P_2}{\Delta t} = -\frac{\Delta P_1}{\Delta t}$$

$$\Delta P_2 = -\Delta P_1$$

$$m_2 v_2 - m_2 u_2 = -(m_1 v_1 - m_1 u_1)$$

$$m_2 v_2 - m_2 u_2 = -m_1 v_1 + m_1 u_1$$

$$m_2 v_2 + m_1 v_1 = m_1 u_1 + m_2 u_2$$

$$2 + 2 = 4$$

$$4 = 2 + 2$$

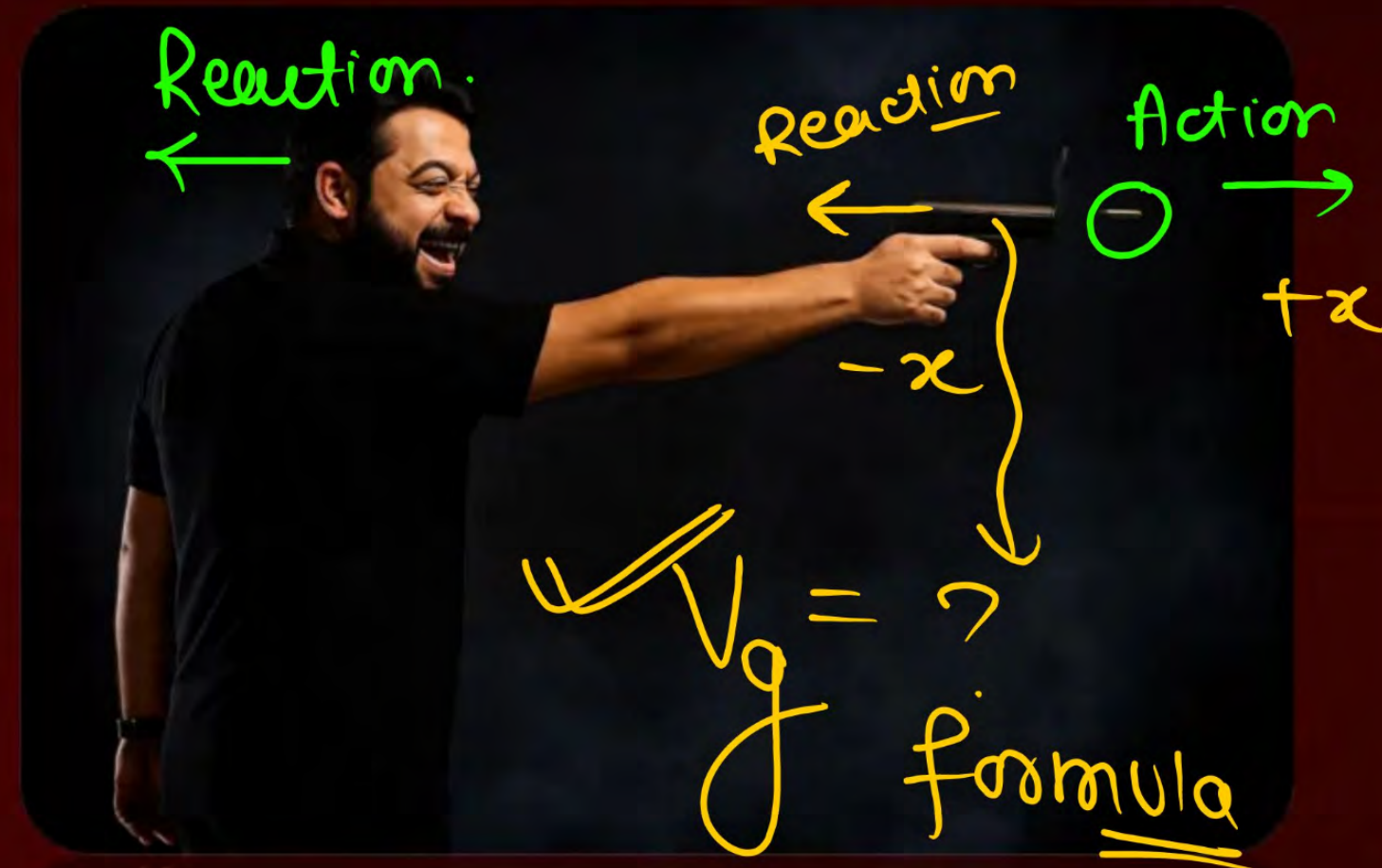
$$\underbrace{m_1 u_1 + m_2 u_2}_{\text{Initial Momentum}} = \underbrace{m_2 v_2 + m_1 v_1}_{\text{Final Momentum}}$$

Application of Law of Conservation of Linear Momentum

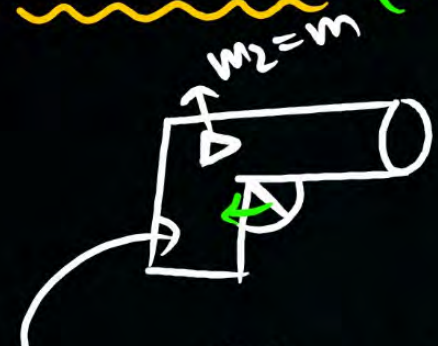


Recoiling of Gun

When a bullet is fired from a gun, the gun moves back which is opposite to the direction of the bullet. This backward motion of gun is called recoil) and the velocity is called as recoil velocity.



Derivation (Recoiling velocity of Gun)



$m_1 = M \rightarrow$ mass of Gun

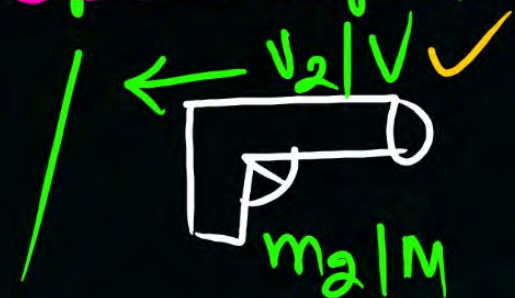
$m_2 = m \rightarrow$ mass of bullet ✓

$u_1 \rightarrow$ initial velocity gun (zero)

$u_2 \rightarrow$ initial velocity of Bullet (zero)

$v_1 = V \rightarrow$ Final velocity of Gun
(Recoiling velocity)

$v_2 = v \rightarrow$ Final velocity of Bullet



$$\Delta m_2/m \rightarrow v_2/v$$

Acc. to law of conservation of momentum



$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

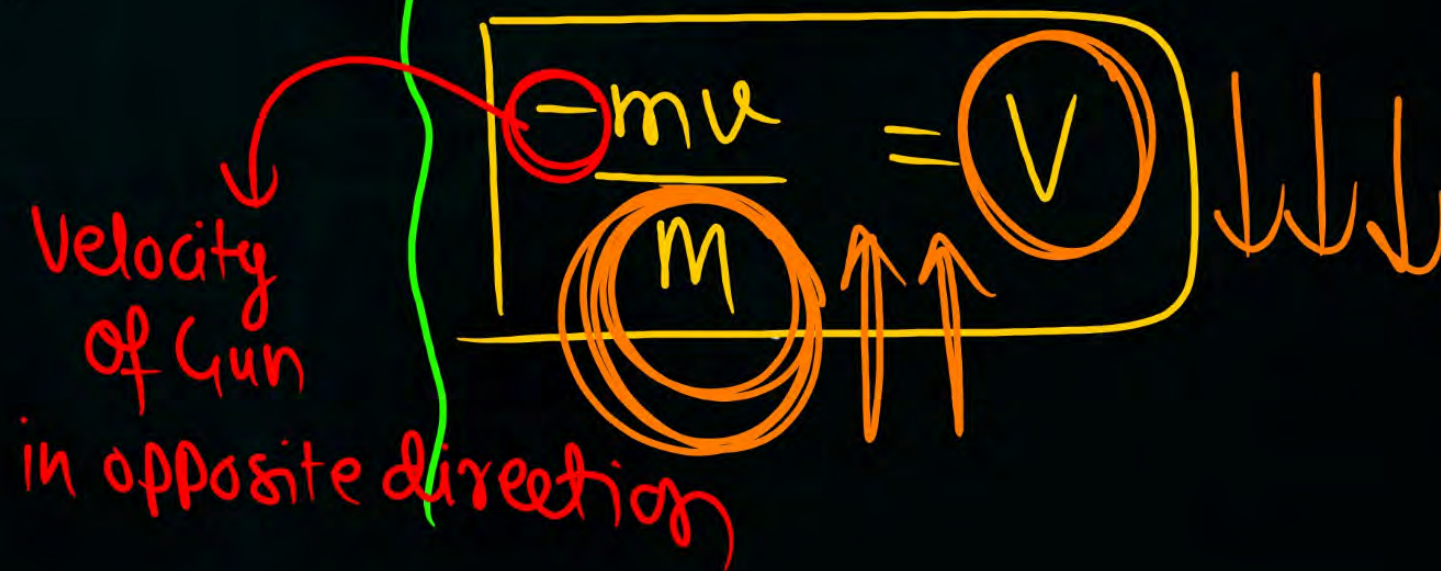
$$m_1 \times 0 + m_2 \times 0 = M V + m v$$

$$0 + 0 = m v + m v$$

$$0 = m v + m v$$

$$-m v = m v$$

Velocity of Gun in opposite direction



$$\downarrow\downarrow V = \frac{\ominus m v}{m \uparrow\uparrow}$$

Q1 Numerical.

Bhat Sahab

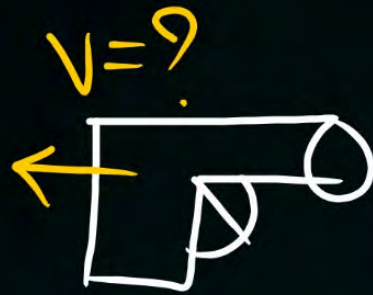

$$m = 1\text{ kg}$$

$$M = 10\text{ kg}$$

Sol

$$V = -\frac{m u}{M}$$

$$V = \frac{-1 \times 100}{10} = -10\text{ m/s}$$



$$100\text{ m/s} \quad (2023)$$

Board

Q.2 Numerical



$F = ?$  $u = 2 \text{ m/s}$ $t = 2 \text{ sec}$  $v = 4 \text{ m/s}$

Shagufta

$m_s = 50 \text{ kg}$ $F = ?$

Sol

$$F = m_s a$$
$$= 50 \times 1$$
$$= \underline{\underline{50 \text{ N}}}$$

$$\# a = \frac{v - u}{t}$$

$$a = \frac{4 - 2}{2} = \frac{2}{2} = 1 \text{ m/s}^2$$



Rocket Launching

2017 JKBOSE

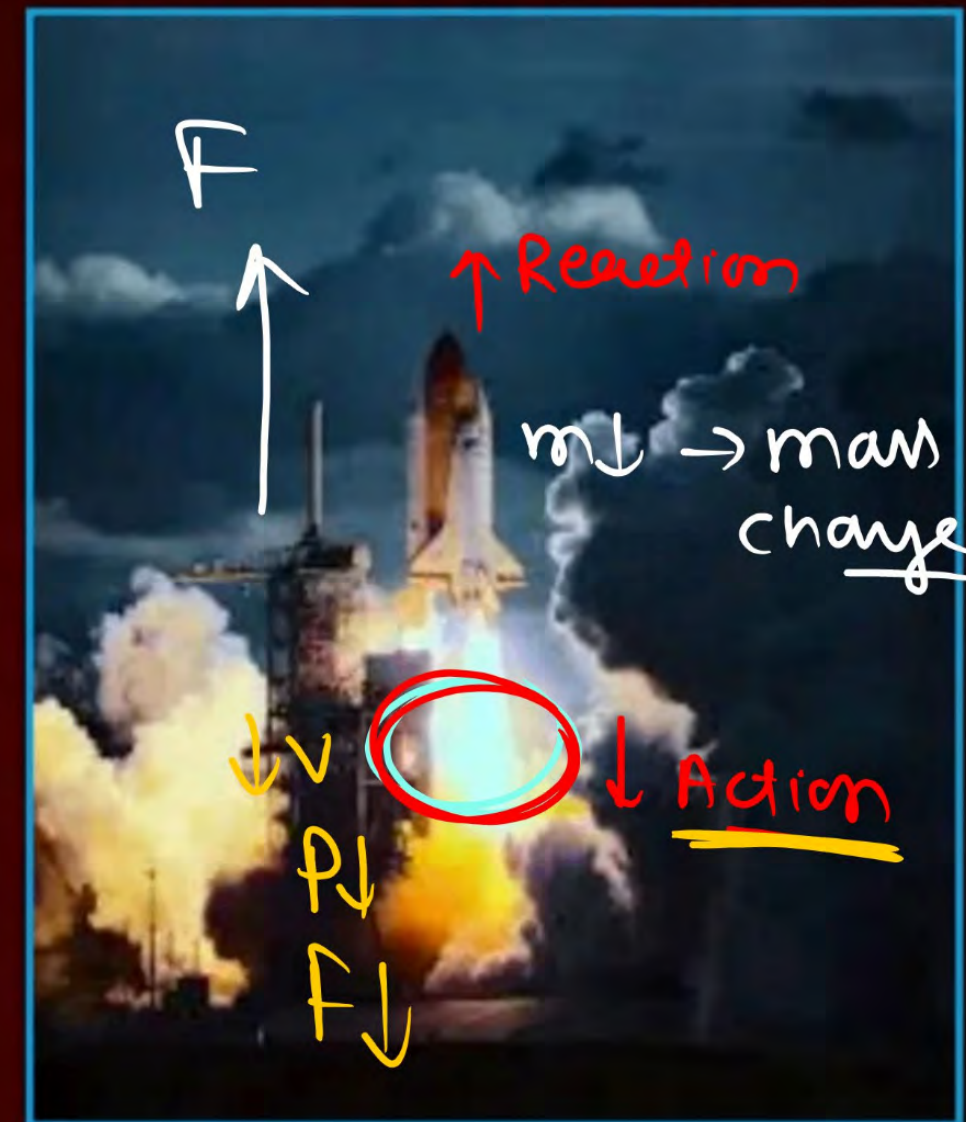


In rocket large amount of gases are produced by burning of suitable fuel within it. When the highly compressed gas escapes at high velocity through the exit in downward direction, it carries large momentum in downward direction. In result to which the rocket moves upward with the same change in momentum as that of the ejected gas.

$$F = ma$$

$$F = m \frac{dv}{dt} + v \frac{dm}{dt}$$

Action = - Reaction ✓

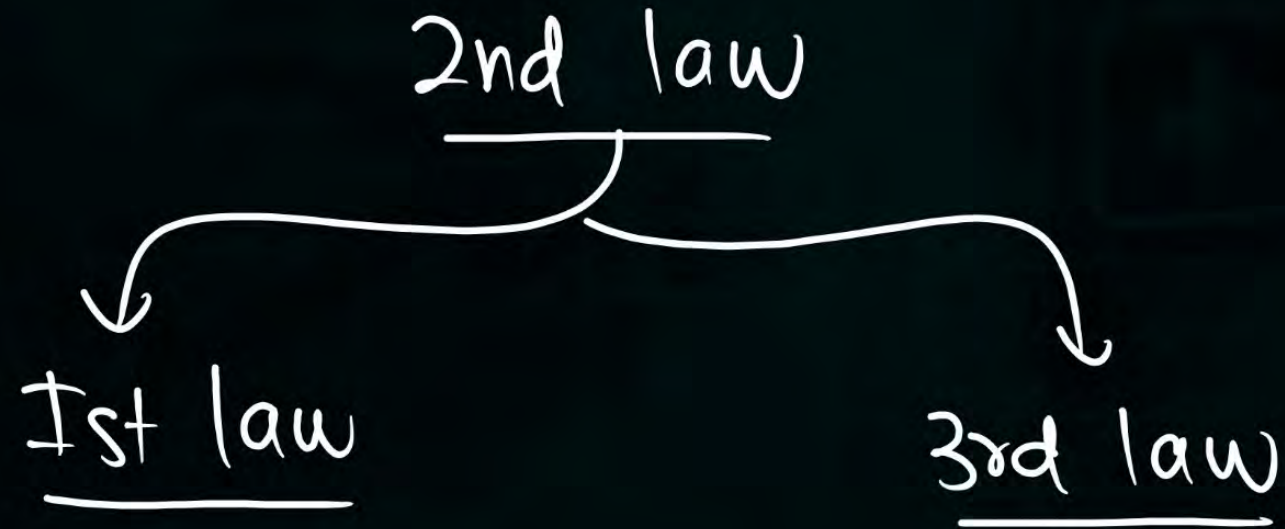


H/W

Q why Newton 2nd law is Klas Real law of motion? or Derive

Ans Because we can derive Both (1st and 3rd law) from 2nd law of Motion

1st and 3rd law from 2nd law of Motion?



(i) 1st law from 2nd law.

$$F = \frac{\Delta p}{\Delta t} \quad \text{2nd law}$$

$$0 = \frac{\Delta p}{\Delta t}$$

$$0 \times \Delta t = \Delta p$$

$$0 = \Delta p$$

$$\Delta p = 0$$

$$p_f - p_i = 0$$

$$p_f = p_i$$

$$\cancel{m}V = \cancel{m}U$$

$$V = U$$

$$\vec{V} = \vec{U}$$

$$0 = 0$$

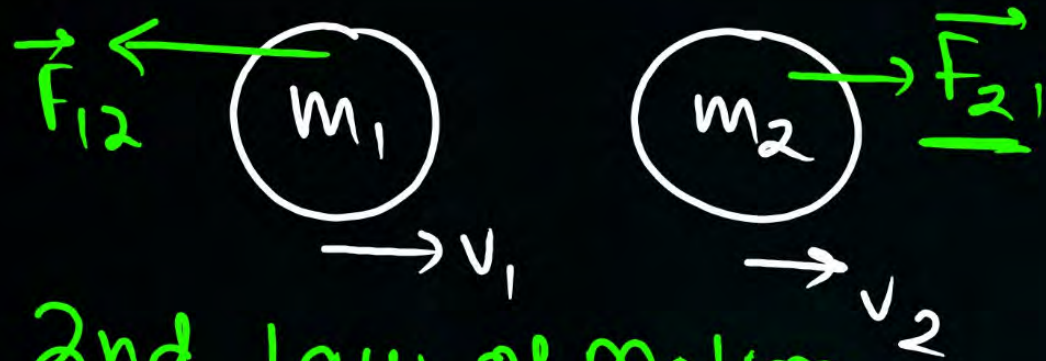
(1st law of Motion)

If the Body is in Rest it will remain in Rest until and unless any external force is applied.

$$F = 0$$



(ii) 3rd law from 2nd law.



2nd law of motion.

$$F_{\text{ext}} = \frac{\Delta p}{\Delta t}$$

$$0 \times \Delta t = \Delta p$$

$$0 = \Delta p$$

$$\Delta p = 0$$

$$\Delta p_1 + \Delta p_2 = 0$$

$$\frac{\Delta p_1}{t} = - \frac{\Delta p_2}{t}$$

$$F_1 = -F_2$$

$$F_2 = -F_1$$

Action = -Reaction

Newton 3rd law of motion



SUMMARY



- ① Newton 3rd law.
- ② law of conservation of momentum.

$$V = -\frac{mv}{m}$$

$$F = m \frac{dv}{dt} + v \frac{dm}{dt}$$

HOMework



Rs -100/-

Q. What are the types of Force? (Discuss) H/u

Thank
You